

Juan P. Morande

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PROFILE

Fifth-year PhD candidate in Industrial and Systems Engineering with expertise in operations research, machine learning, optimization, and simulation. Proven background in solving complex, research-driven problems in logistics, supply chain, scheduling, and forecasting. Proficient in programming for analytical and computational modeling. Experienced in leading teams and working independently, with strong verbal communication skills demonstrated through conference presentations and teaching. Seeking an applied research engineering role at the intersection of operations research and machine learning, focused on optimization, predictive modeling, and ML-augmented decision systems.

EDUCATION

- **PhD, Industrial and Systems Engineering** | Auburn University | August 2021 – December 2026 (expected)
- Dissertation: *Integrating Prediction and Optimization: Data-Driven Methods for Complex Logistics Problems*.
- Advisors: Alice E. Smith, Daniel Silva
- **MS, Industrial and Systems Engineering** | Auburn University | May 2023
- **BS, Industrial Engineering** | Universidad de los Andes, Chile | October 2020
- Thesis: *Benchmarking statistical techniques and optimization heuristics applied to discrete-event simulation*.

RECENT EXPERIENCE

RESEARCH AND TEACHING ASSISTANT | AUBURN UNIVERSITY | AUGUST 2021 – PRESENT

- Developed optimization models and heuristics for aircraft and crew allocation and scheduling to support wargaming and planning (AFMC, Frontier Technology Inc., 2023–2025). Delivered optimization modules (MILP and metaheuristics solvers) for large-scale instances.
- Conducted research on deep learning and reinforcement learning applications in illicit supply chains, focusing on opioid precursor trafficking patterns (DHS CAO Grant, 2024–2025).
- Designed hierarchical optimization methods for a two-dimensional vehicle loading problem with heterogeneous fleets, minimizing delays in an industrial application, improving a daily dispatch metric by 27%. Submitted to European Journal of Operational Research.
- Applied explainable AI methods (over LSTM, Transformers, TFT) to forecast freight rate dynamics in Chilean ports, extracting actionable insights into rate drivers from the top-performing models (6%-12% MAPE). Submitted to Annals of Operations Research.
- Led applied projects covering discrete-event simulation of a Chilean mining operation, a hybrid RAG system over legal documents for a Chilean law firm, LLM-based agentic simulations, and memory-enhanced agentic framework for combinatorial optimization.
- Teaching Assistant for Engineering Economics, Statistical Quality Control, Data Mining, and Adaptive Optimization.

VISITING RESEARCHER | NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY | FEBRUARY 2024 – MAY 2024

- Developed closed-loop supply chain MILP models to evaluate strategic and product design decisions using scenario-based analysis. Published in *Omega* (2025).

SKILLS

- Mathematical optimization (MILP, stochastic, robust), scheduling and routing problems (Pyomo, Gurobi, CPLEX)
- Statistical methods, experimental design, Bayesian modeling (hierarchical, MCMC), simulation (DES, ABM)
- Machine learning (PyTorch, decision-focused learning), time-series (LSTM, Transformers, TFT), XAI (LIME, SHAP), Reinforcement learning (Q-learning, DQN, PPO)
- LLM applications (RAG, multi-agent coordination, memory architecture), fine-tuning (QLoRA, RLHF)
- Programming: Python (primary), R, Java, SQL. Languages: Native Spanish and fluent English

AWARDS & INVOLVEMENT

- Institute of Operations Research and Management Sciences (INFORMS), local chapter | Secretary (2022-2023), Vice-president (2023-2024), President (2024-2025), Vice-president (2025-2026), E-council representative (2026-present).
- SPEC Cybersecurity Program Scholarship, UVA at Wise (2019), Amazon-Scot INFORMS scholarship (2022)
- Multiple conference presentations: INFORMS Annual Meeting (2022, 2023, 2025), IFORS (2023), OPTIMA (2025), ICPR Americas (2020)